Weixin Chen

Ph.D. Candidate @ University of Illinois Urbana-Champaign

CONTACT INFORMATION

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RESEARCH INTERESTS

- **Trustworthy Machine Learning:** Interpretability, Robustness, Generalization
- **Trustworthy Large Models:** Truthful Generation, Robust Reasoning
- **Neuro-Symbolic AI.**

EDUCATION

University of Illinois Urbana-Champaign — *Urbana, USA*
Ph.D. in Computer Science — *Aug. 2023 - Present*
Advisor: Prof. Han Zhao
GPA: 4.0/4.0

Tsinghua University — *Beijing, China*
M.E. in Artificial Intelligence — *Aug. 2020 - Jun. 2023*
Advisor: Prof. Haoqian Wang
GPA: 4.0/4.0

Sun Yat-sen University — *Guangzhou, China*
B.S. in Mathematics — *Aug. 2016 - Jun. 2020*
GPA: 4.0/4.0

PROFESSIONAL EXPERIENCES

Simons Institute, UC Berkeley — *Berkeley, USA*
Visiting Student — *Aug. 2024 - Dec. 2024*
Modern paradigms in generalization.

Secure Learning Lab, UIUC — *Urbana, USA*
Research Intern *(Advisor: Prof. Bo Li)* — *Jul. 2022 - Jan. 2024*
Backdoor attacks on diffusion models.
Post-processing for LLM truthfulness.

SCLBD, CUHK-Shenzhen — *Shenzhen, China*
Research Assistant *(Advisor: Prof. Baoyuan Wu)* — *Jun. 2021 - May. 2022*
In-training backdoor defenses.

PUBLICATIONS

Weixin Chen, Antonio Vergari, Han Zhao. "Can VLMs Reason Robustly? A Neuro-Symbolic Investigation". *Under Review*, 2025.

Weixin Chen, Han Zhao. "Understanding and Improving Adversarial Robustness of Neural Probabilistic Circuits". In *Proceedings of the 39th Advances in Neural Information Processing Systems* **(NeurIPS)**, 2025.

Weixin Chen*, Simon Yu*, Huajie Shao, Lui Sha, Han Zhao. "Neural Probabilistic Circuits: Enabling Compositional and Interpretable Predictions through Logical Reasoning". *Under Review (arXiv:2501.07021)*, 2025.

Weixin Chen*, Simon Yu*, Huajie Shao, Lui Sha, Han Zhao. "Neural Probabilistic Circuits: An Overview". In *the 8th Workshop on Tractable Probabilistic Modeling at UAI* **(TPM)**, 2025.

Weixin Chen, Dawn Song, Bo Li. "GRATH: Gradual Self-Truthifying for Large Language Models". In *Proceedings of the 41st International Conference on Machine Learning* **(ICML)**, 2024.

Boxin Wang*, **Weixin Chen***, Hengzhi Pei*, Chulin Xie*, Mintong Kang*, Chenhui Zhang*, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, Zinan Lin, Yu Cheng, Sanmi Koyejo, Dawn Song, Bo Li. "DecodingTrust: A Comprehensive Assessment of Trustworthiness in GPT Models". In *Proceedings of the 37th Advances in Neural Information Processing Systems* **(NeurIPS)** **(Oral, Outstanding Paper Award)**, 2023.

Weixin Chen, Dawn Song, Bo Li. "TrojDiff: Trojan Attacks on Diffusion Models with Diverse Targets". In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* **(CVPR)**, 2023.

Weixin Chen, Baoyuan Wu, Haoqian Wang. "Effective Backdoor Defense by Exploiting Sensitivity of Poisoned Samples". In *Proceedings of the 36th Advances in Neural Information Processing Systems* **(NeurIPS)** **(Spotlight)**, 2022.

TEACHING EXPERIENCE

Teaching Assistant, UIUC

CS 442: Trustworthy Machine Learning — *Spring 2025*

Teaching Assistant, Tsinghua University

Modern Signal Processing — *Spring 2022*

HONORS & AWARDS

AISTATS Best Reviewer	*2025*
NeurIPS Outstanding Paper Award	*2023*
Wing Kai Cheng Fellowship, UIUC	*2023*
First Prize Scholarship (top 3%), Tsinghua University	*2022*
First Prize Scholarship (top 5%), Sun Yat-sen University	*2017, 2018, 2019*
National Scholarship (top 2%), Sun Yat-sen University	*2018, 2019*
First Prize (top 1%), Chinese Mathematics Competitions (CMC)	*2018*
Giordano Donation Scholarship (top 3%), Sun Yat-sen University	*2017*

SERVICES

Journal Reviewer

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
IEEE Transactions on Information Forensics & Security (TIFS)
Transactions on Machine Learning Research (TMLR)

Conference Reviewer

AAAI Conference on Artificial Intelligence (AAAI)	*2026*
Advances in Neural Information Processing Systems (NeurIPS)	*2024, 2025*
International Conference on Machine Learning (ICML)	*2024, 2025*
International Conference on Artificial Intelligence and Statistics (AISTATS)	*2025, 2026*
International Conference on Learning Representations (ICLR)	*2025, 2026*

SKILLS

Programming: Python, PyTorch, LaTeX
Languages: English *(fluent)*, Chinese *(native)*, Cantonese *(native)*